HO ZHU YI

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ACADEMIC QUALIFICATIONS

London, UK — **Imperial College London** — **Oct 2015 – Jun 2019**

- **Major:** MEng in Civil Engineering (First Class Honours 87%)
- **Awards:** Dean's list recipient (top 10% in years 1-4), ICE Prize (best in third and fourth years), Letitia Chitty Prize (greatest merit in the third year), Peter Fraenkel Prize (best third year Group Design Project)
- **Programming Courses:** Computational Methods 1 & 2, Operational Research and Systems Analysis

PROFESSIONAL EXPERIENCE

Computational Structural Engineer — **Arup Jururunding Sdn. Bhd. / Arup Singapore Pte. Ltd.** — **Aug 2019 – Sep 2025**

As a key engineer involved in large-scale building projects, I develop computational models and workflows for structural analysis, optimisation and design in Python, VBA, and Grasshopper (visual programming tool).

- Built Python API wrappers that enables data engineering pipelines that interoperates between multiple commercial structural analysis tools, abstracting low-level .NET calls into clean interfaces and streamlining multi-tool engineering workflows.
- Standardized structural models into unified Polars DataFrame schemas, ensuring consistent data loading and transformation.
- Engineered Python libraries that combine numerical optimisation, international code-compliant design, and high-performance data processing with NumPy and Polars. Delivered scalable workflows and reusable modules that automated reinforcement optimisation of columns, piles, walls, and coupling beams, enabling validated analysis across large structural models.
- Implemented high-performance algorithms (e.g. ray–triangle intersections) with NumPy and Numba to accelerate analysis by 100× compared to commercial tool, supporting large-scale model evaluation.
- Designed a Python logging framework integrated with Jupyter Notebooks that generates automated Altair visualisations, enabling consistent tracking, comparison, and reporting of structural analysis models and results.
- Created reproducible Jupyter Notebook workflows for solution exploration and communication with interactive visualisations.

In addition to technical contributions, I am well versed in the management of major building projects and their design objectives.

- *(M'sia):* Merdeka 118 Tower, KLCC Lot LL1M, IOI ICM3 Iconic Tower, SDB Melawati; *(S'pore):* Toa Payoh Integrated Development.
- Liaised as the main engineer with external multidisciplinary consultants to deliver a total design.
- Led and managed internal teams of engineers from Shang Hai, Kuala Lumpur and Singapore offices to complete detailed design towards a successful tender delivery.

RELEVANT PROJECTS

My personal/professional projects that involve programming, data analysis and backend development; written in line with clean code principles and coding guidelines (e.g. PEP8).

- **Code & Conundrums (2025-):** A blog where I share interesting problems and the code-driven journeys behind solving them [Link]:
  - **Mapping the Milky Way:** Leveraged Python (Polars, Astropy, Altair) to perform high-efficiency data processing, analysis, and visualization of Gaia DR3 stellar measurements, revealing structural patterns of the Milky Way.
  - **High-Performance SEO Blogging:** Developed a workflow integrating Quarto-generated Jupyter Notebook HTML with the Astro framework to create high-performance, SEO-friendly blogs.
- **Movie Recommender (2025):** React + TS + AWS stack, featuring: auth'n and auth'z with Cognito, compute / storage with Go + Lambda + DynamoDB, infrastructure as code and continuous deployment with CloudFormation + Gitlab CI/CD. [Link, Front, Back]
- **Structural Engineering Library (2020-2025):** A compilation of Python libraries for the developed workflows.
- **Individual Research Project (2019, 86%):** Simulated filtration processes using a flow-biased walk simulation of soil particles through realistic void spaces represented as networks. Utilised Matlab, Git, and object-oriented programming (OOP). [Link]

PROFESSIONAL CERTIFICATIONS

Full Stack Development with AI (99.8%) — **National University of Singapore** — **Apr – Oct 2025**

- Frontend development with JavaScript/TypeScript (React, Vite, Tanstack Router + Query, Zod, Zustand, Shadcn, Tailwind).
- Backend development with JavaScript/TypeScript (Express.js, Pug, Sqlite3) and Python (Flask, Jinja2, Sqlite3).
- SQL and NoSQL database management with MySQL, PostgresSQL, MongoDB.
- Machine learning and artificial intelligence with Python (NumPy, Pandas, Matplotlib, Seaborn, Scikit-Learn, TensorFlow, MLFlow).

CS50 Intro to Computer Science — **Harvard University** — **Apr – Oct 2025**

- Programming, data structures and algorithms with C; frontend dev with HTML, CSS, JS; backend dev with Python (Flask), SQL.

SKILLS

Language: *(fluent):* English, Mandarin, Malay.
Programming: *(proficient):* HTML, CSS, JavaScript, TypeScript, Python, Go, C, SQL, NoSQL, VBA, Matlab, Git, Bash.
Backend: AWS (Amplify, SDK for JS / Go, CloudFormation, Cognito, Lambda, DynamoDB), RESTful API, GitLab CI/CD.
Others: Maths, Statistics, Data visualisation, Office suite, Project management.